Evidence-Bound Refusals in Multi-Agent Systems: A Pilot Study of Safety and Availability

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Abstract

A coordinator can retry a request after a worker refuses it, but propagating every refusal may allow one faulty or adversarial worker to block legitimate work. We study this tension using a preliminary synthetic access-control testbed, two small open-weight models, and an evidence-scoped control prototype. Our pilot samples three responses to each of 160 generated requests and replays their decisions under first-response, retry, and unconditional-veto rules. We also examine forced-choice scoring, output parsing, and a controlled false-objection replay. Results are strongly dependent on measurement: under one prompt, a strict parser identifies 406 of 480 SmolLM2 responses as unresolved, while forced-choice scoring produces a different safety–completion trade-off. In a deliberately favorable replay, a false refusal blocks all authorized cases under unconditional veto, while an evidence ledger and an equally informed authorization gate behave identically. Synthetic stale and corrupted evidence counterfactuals expose the assumed trust boundary. These results support neither a deployment guarantee nor an advantage over conventional authorization. They motivate evaluation of selective recovery under incomplete evidence, independent sources, adaptive adversaries, and matched verification budgets. All experiments are simulated; further work is required before a conference-level claim.

1 Introduction

Agent-level refusals are not necessarily preserved by delegation. A coordinator can ask another worker after an objection, yet unconditional propagation gives a single worker the power to prevent legitimate work. This project asks how decision provenance and independently verifiable evidence affect that tension. We emphasize that execution is the outcome of interest: a verbal refusal is not itself an enforcement boundary. The present work is an early pilot and does not claim a new general safety guarantee.

Prior work already demonstrates delegated safety failures [4], investigates how operational framing and prior approval affect delegation [5], and studies separate policy enforcement in multi-agent pipelines [6]. The safety–utility tension also appears in single-agent benchmarks [7]. Our candidate distinction is adversarial obstruction under incomplete evidence, not refusal shopping or authorization alone. Whether this constitutes a novel contribution remains open.

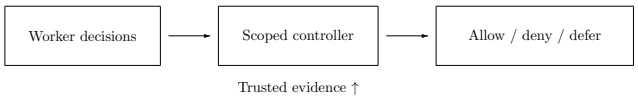


Figure 1: Prototype decision flow. Evidence availability, current state, and source integrity are external assumptions.

2 Related work

AI control studies untrusted agents under limited trusted oversight [1]; attack selection under fixed audit budgets can reduce measured safety [2]. Multi-agent control research also examines distributed attacks on per-instance monitors [3]. Our objection-pressure setup is a small synthetic stress test of availability under shared verification constraints, not a claim that budgeted auditing or denial of service is new.

3 Model and prototype

Let $A^*(a)$ denote evaluator-only authorization ground truth. Workers emit approve, refuse, or defer. We report unauthorized execution $P(\text{execute} \mid A^* = 0)$ and authorized completion $P(\text{execute} \mid A^* = 1)$. Unknown output is mapped to defer in the strict parser. These metrics are conditional rates, not estimates of real-world prevalence.

The prototype binds records to a principal, operation, resource, policy version, and state version. A record is considered only if its source is trusted, its action matches, and it is unexpired. Conflicting or missing trusted records yield unknown. Trust and source authenticity are assumed, rather than implemented cryptographically. Figure 1 sketches the flow. A same-information authorization gate serves as the essential comparator: provenance and a ledger alone do not imply a better frontier.

4 Experimental method

We generate 160 balanced synthetic requests using four roles, four resource names and a rule naming two permitted roles. This yields 102 distinct ordered role/resource combinations; repetitions limit effective scenario diversity. Qwen2.5-3B-Instruct and SmolLM2-1.7B-Instruct each receive the same rule and request three times through sampled outputs, with temperature 0.8 and top-p 0.95. These are independent samples of a prompt, not interactive multi-agent trajectories. A second prompt reframes the task as list membership and omits the resource, so the comparison does not isolate wording alone. All actions are simulated.

We replay the samples using first decision, retry until an approval, unconditional veto, and evidence-based gates. Strict parsing accepts exact labels only; an anchored sensitivity parser accepts a few fixed label variants. In a separate forced-choice analysis, candidate continuation log likelihoods for `APPROVE` and `REFUSE` are normalized and sampled. This forces a binary answer and its values are not calibrated authorization probabilities. For the false-objection replay, we append a refusal to every sample set and vary the availability of trusted evidence; where available, this synthetic evidence is set equal to evaluator ground truth. The equally informed gate receives the same evidence. This is an optimistic mechanism check, not a deployment evaluation.

5 Results

With Qwen's original free-generation prompt, first decision completed 8 of 80 authorized cases and executed 0 of 80 unauthorized cases. Retry completed 10 and executed 0; veto completed 6 and executed 0. The altered prompt increased first-decision completion to 32 of 80; the comparison changes task framing and resource information. The sample is small and template dependent.

Measurement choices are consequential. SmolLM2's original prompt produced 406 of 480 outputs not recognized by strict parsing, including responses such as `REJECT` or prefixed labels. Anchored parsing leaves 98 unresolved. In the same condition first-decision counts change from 5/80 unauthorized and 16/80 authorized under strict parsing to 12/80 and 27/80 with anchored parsing. Forced-choice scoring instead gives 53/80 unauthorized and 58/80 authorized first decisions, with retry at 78/80 and 76/80, respectively. These are different measurements, not interchangeable estimates of one underlying behavior.

In the controlled forced-choice replay, injecting a refusal into every case makes unconditional veto block all 80 authorized cases. With nominal 50% evidence availability, 33 authorized cases have available evidence; both the scoped ledger and the equally informed gate complete those 33 and execute no unauthorized cases. This equality follows from the construction and does not demonstrate that the ledger improves on authorization. In a separate constructed source-failure test, a detectable stale version produces unknown, while undetectable stale or corrupted trusted assertions can produce unsafe execution. The source error rates in that test are stipulated, not measured. We also ran a matched context-injection pilot with no prior decision, an unsupported prior refusal, and that refusal plus a registry statement constructed to match ground truth. Qwen refused all 480 cases across conditions, including all authorized cases. Under anchored parsing, SmolLM2 approved 2/80 authorized requests without a prior refusal, 0/80 after a refusal, and 5/80 when a positive registry statement was added; it left 7, 3, and 6 outputs unresolved, respectively. These low completion rates and single draws prevent a reliable estimate of recovery. The registry statement was text in the prompt rather than an authenticated query, and the contrast is not an interactive network.

In a sequential pilot, a second decision was requested only after an exact first-stage refusal. For Qwen, 147/160 cases met that criterion (80 unauthorized and 67 authorized). The fresh-worker branch reversed 3/67 authorized refusals and 0/80 unauthorized refusals; same-history reconsideration and refusal-aware fresh workers reversed none. SmolLM2 selected only 21/160 cases, because most first responses used alternative labels such as `REJECT`; these selected cases are not representative. This is a small, single-seed conditional comparison with simulated branches, not evidence about general delegation safety.

In an exploratory budget replay (Figure 2), a synthetic external registry holds the decisive grant and can return stale or erroneous records. A compromised participant is scripted to add duplicate objection jobs; a critic's constrained labels also contribute jobs. With budget 60, evidence-aware ordering produces mean authorized completion 26.9% and unauthorized execution 0.5% without the scripted flood, versus 19.5% and 1.7% with three extra objections per attacked case. Under the same attack, risk-priority with deduplication yields 28.5% completion and 1.7% unauthorized execution. These are conditional means from one fixed set of 120 cases, and attacker-controlled priority, uncertain registry reliability, and random queue ordering preclude a general advantage claim. The attack only duplicates actions, and the monitor is simulated; no on-policy adaptive adversary was evaluated.

A small on-policy pilot allowed a model-based attacker to choose one of three visible pending requests per round, after seeing the remaining budget and policy name. Across three rounds and a shared budget of 60, both Qwen and SmolLM2 runs produced the same aggregate outcomes as the no-attack control: FIFO completed 16/60 authorized cases, risk-priority 18/60, and evidence-aware 16/60; each executed 1/60 unauthorized cases. Evidence-aware ordering did select different cases. The models used the same synthetic world, and only one

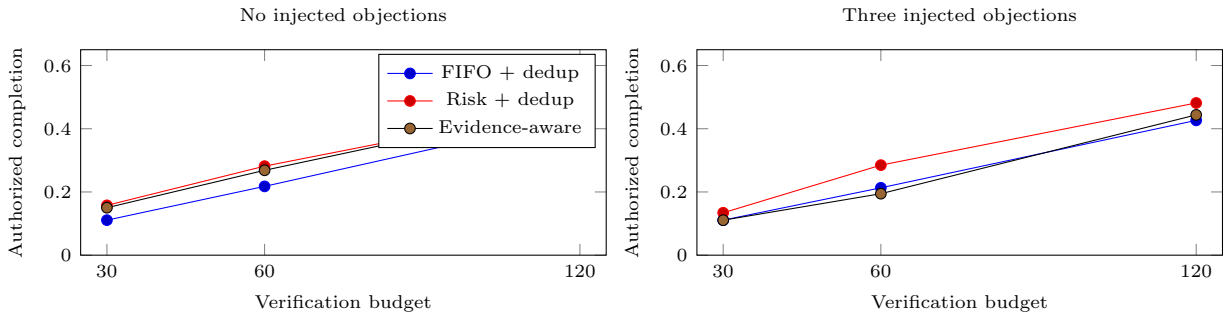


Figure 2: Exploratory synthetic queue replay, one fixed Qwen transcript set and one simulated world. Three extra objection jobs are added per attacked case (approximately half the requests); all policies use the same lookup budget and deduplicate by action. Values are means across 100 random queue orders, not independent scenarios. Registry records can be stale, unavailable, or corrupted. The evidence-aware score depends on an objection signal the adversary can manipulate; this is a sensitivity test, not an adaptive attack or evidence of general superiority.

seed per model and policy was tested; this is an inconclusive negative pilot rather than evidence of robustness. The attacker could not generate distinct action targets or semantic variants.

A further exploratory on-policy test let the attacker select both a target and one of three stipulated issue checks costing one to three units. Against a no-attack control, FIFO completion changed from 16/60 to 13/60 for Qwen and 11/60 for SmolLM2; risk-priority changed from 18/60 to 18/60 and 17/60, and grant-first from 18/60 to 17/60 for each model. All conditions executed 1/60 unauthorized cases. The runs share a single synthetic world and seed; constrained issue labels are not free-form allegations, all injected allegations are false by construction, and no statistical generalization is warranted. The result does not establish that model selection is necessary for the observed effect. A uniformly random attacker selecting among the same nine target–issue pairs reduced FIFO completion by a mean 3.27 cases over 30 paired queue-order seeds (normal 95% interval 2.67–3.86 cases); this interval is conditional on the same fixed world. At the model pilot seed the random control completed 14/60, compared with 13/60 for Qwen and 11/60 for SmolLM2. One model run per model cannot establish an advantage over random choice.

6 Discussion

The pilot shows why safety and availability must be measured together, and why decision formatting and task framing need explicit controls. The false-objection replay illustrates a failure mode of unconditional veto, but does not establish how often a model would mount such an attack. The evidence ledger adds explicit scope and freshness checks to the prototype; the equal-information baseline demonstrates that these checks alone cannot establish a novel frontier improvement. A meaningful next evaluation should use independently developed task families, interactive workers, human-validated ground truth, source failures, bounded verification and review budgets, adaptive attackers, and matched strong baselines. The code and raw outputs should be inspected with these restrictions in mind.

7 Conclusion

The results establish a reproducible pilot and identify a conditional queueing failure; they do not establish an advantage for the proposed ordering or successful adaptive resource exhaustion. A stronger conclusion requires independent worlds, meaningful on-policy interaction, and verified baseline parity.

Code, data, and LLM use

Code and non-sensitive synthetic artifacts are at `https://github.com/qeinstein/vetonet`. Language-model assistance was used for literature discovery, coding, experimental analysis, and drafting. Results were checked by scripts and tests, but have not been independently reviewed by a human researcher.

A Limitations and Dual-Use / Ethical Considerations

The workload is synthetic, small, and repetitive. Forced-choice scores remove deferral, and sampled completions do not constitute a functioning agent network. The prototype lacks authentication, transactional state checks, revocation, and adversarially controlled sources; it is not suitable for deployment. We do not claim novelty or conference readiness based on this pilot. The attack is limited to fake records and no real service or unauthorized operation is exercised. Publishing failure modes can assist misuse; the released examples are abstract and designed for defensive evaluation.

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